AlignLight: Spatiotemporal Alignment via Geometric Modulation for Traffic Signal Control

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Abstract

Graph neural networks (GNNs) have shown potential in multi-agent reinforcement learning (MARL) for traffic signal control. However, many existing node-centric architectures rely on static graph topologies, which often limits their generalization when applied to unseen road networks. Furthermore, these methods typically treat road segments as binary connections and perform information aggregation based on single-step snapshots, failing to explicitly account for the physical propagation delay of traffic flow. To address these limitations, we propose AlignLight, an attention-based framework that integrates continuous geometric coordinates to capture spatiotemporal physics. First, we employ a sequence encoder to extract macroscopic temporal features from historical traffic data. To model traffic wave propagation, we design a spatiotemporal attention mechanism that utilizes geometric distances to modulate temporal attention weights, explicitly aligning the model's focus with the physical travel time of vehicles. Finally, we formulate an ego-decoupled graph aggregation strategy that incorporates continuous spatial coordinates directly into the message-passing process, eliminating topological confounding from self-connections. Extensive experiments on real-world large-scale benchmarks demonstrate that AlignLight: (1) stabilizes decentralized multi-agent training dynamics; (2) achieves competitive optimization efficiency against both topological GNNs and recent sequence-modeling baselines; and (3) exhibits superior zero-shot transferability across unseen, heterogeneous road topologies.

Introduction

Multi-agent reinforcement learning (MARL) for urban traffic signal control (TSC) has demonstrated potential in optimizing complex networked systems (Wei et al. 2020). To enable scalable coordination, integrating graph neural networks (GNNs) has become standard practice, facilitating message passing and network-level cooperation among decentralized agents (Chu et al. 2020; Wei et al. 2019b). By representing intersections as nodes and road segments as edges, these architectures aim to learn collective policies that harmonize local decisions for global efficiency.

However, the generalization capability of many GNN-based architectures is often constrained by a reliance on static graph topologies. Standard graph attention mechanisms (Veličković et al. 2018) typically treat road segments as binary connections or static weights, overlooking the fact that physical roads dictate specific spatial distances and directional flow between agents. When models are trained on specific graph structures, they risk overfitting to these memorized connectivity patterns. Consequently, these architectures exhibit performance degradation when evaluated on road networks with different topologies (out-of-distribution shifts).

Furthermore, prevalent approaches often rely on single-step snapshots, which may limit their capacity to capture the dynamic state propagation inherent in traffic networks. To address this, sequence-modeling architectures, such as TransformerLight (Wu et al. 2023) and Decision Transformer (Chen et al. 2021a), have been introduced to capture long-range temporal trends. However, these sequential frameworks frequently project spatiotemporal features into unaligned high-dimensional vectors or decouple the spatial and temporal dimensions entirely. Consequently, they do not explicitly model a fundamental physical phenomenon: *propagation delay*. In practice, traffic flow from neighboring intersections reaches a target intersection with a temporal offset determined by the physical distance between them, a kinematic property that pure sequence-matching architectures typically omit.

To mitigate these interrelated limitations of static topological overfitting and unaligned temporal propagation, we propose **AlignLight**, an online decentralized MARL framework that integrates continuous geometric coordinates into an attention-based architecture to model spatiotemporal physics. AlignLight comprises three core components. First, it employs a sequence encoder to extract temporal features from historical traffic observations. Second, we design a spatiotemporal attention mechanism that explicitly models physical propagation delays. By utilizing the geometric distance between intersections to modulate temporal attention weights, the model dynamically aligns its focus with the physical travel time of traffic flow. Finally, we formulate an *ego-decoupled graph aggregation* strategy that incorporates continuous spatial coordinates into the message-passing process. Unlike general-purpose graph attention variants (e.g., GATv2 (Brody, Alon, and Yahav 2021)) that enforce self-loops to stabilize node representations, our strategy isolates the internal ego-state from neighbor pooling, enforcing an information bottleneck that eliminates topological confounding from self-connections.

Our primary contributions are summarized as follows:

- We propose AlignLight, an attention-based framework for MARL-based TSC that integrates continuous geometric coordinates to model spatiotemporal physics. By generating dynamic temporal shifts based on spatial coordinates, it explicitly aligns the model's focus with the physical propagation of traffic waves.
- We formulate an ego-decoupled graph aggregation strategy utilizing continuous spatial encodings. By separating ego-states from neighbor features during message passing, this approach minimizes topological confounding and enhances zero-shot transferability across unseen networks.
- Extensive experiments on real-world large-scale benchmarks demonstrate that AlignLight stabilizes multi-agent training and achieves competitive performance, outperforming both topological GNN baselines and recent sequence-based Transformers (e.g., TransformerLight). Furthermore, it exhibits robust generalization capabilities across unseen road topologies with minimal computational overhead at the millisecond scale.

Related Work

MARL in Traffic Signal Control

Traffic signal control has historically relied on pre-timed heuristics or rule-based adaptive systems, which often struggle to scale under highly dynamic traffic fluctuations (Wei et al. 2020). The advent of deep reinforcement learning introduced data-driven paradigms, with pioneering works like IntelliLight (Wei et al. 2018) demonstrating the efficacy of DQN at isolated intersections. To coordinate network-level traffic, MARL has been widely adopted (Chu et al. 2020). Extensive efforts have focused on enhancing state and reward representations; for instance, PressLight (Wei et al. 2019a) and Advanced-XLight (Zhang et al. 2022) incorporated the max-pressure principle to optimize vehicle distributions.

However, these methods typically model the environment using single-step snapshots. Because urban traffic states are inherently partially observable, relying on instantaneous snapshots limits the capacity to capture underlying system dynamics. While certain approaches attempt to mitigate this limitation via recurrent neural networks (RNNs), they often exhibit vulnerability to long-horizon credit assignment challenges. This representation bottleneck is addressed in our framework by integrating a sequence encoder that compresses historical observations to extract macro-level features while filtering localized observation noise.

GNNs for Spatial Coordination

In multi-agent systems, an agent's policy optimization heavily depends on the states of its surrounding neighbors. Consequently, GNNs have emerged as a standard tool to capture spatial dependencies within road networks (Battaglia et al. 2018). A prominent milestone in TSC is CoLight (Wei et al. 2019b), which applies graph attention networks (GATs) (Veličković et al. 2018) to learn dynamic communication weights among intersections. Despite these advancements, standard node-centric GATs compute attention weights based on predefined discrete adjacency matrices, which risks causing the model to overfit to memorized topological connectivity patterns. Furthermore, these architectures typically treat physical roads merely as binary connections, omitting the continuous spatial distances between intersections.

While relative spatial encodings have been explored in general graph or language models (Ying et al. 2021; Shaw, Uszkoreit, and Vaswani 2018; Chu et al. 2021), they generally map discrete indices or grid intervals within a structural latent space. In contrast, AlignLight maps continuous, non-grid physical coordinates directly into the graph message-passing process. Combined with an ego-decoupled graph aggregation strategy that isolates ego-states from neighbor features, our framework ensures that information routing is governed by physical geometric spatial distributions rather than fixed discrete topologies, thereby preventing topological over-smoothing and enhancing out-of-distribution robustness.

Sequence Modeling on Dynamic Graphs

Traffic flow is inherently a time-series process evolving over a spatial graph topology. Early sequential methods integrated temporal convolutions with GNNs to model these combined patterns (Zhang et al. 2020; Fang et al. 2020). With Transformer architectures (Vaswani et al. 2017) expanding sequence-modeling capabilities, paradigms like TransformerLight (Wu et al. 2023) introduced gated transformers to TSC, while frameworks like Decision Transformer (Chen et al. 2021b) recast reinforcement learning as conditional sequence generation.

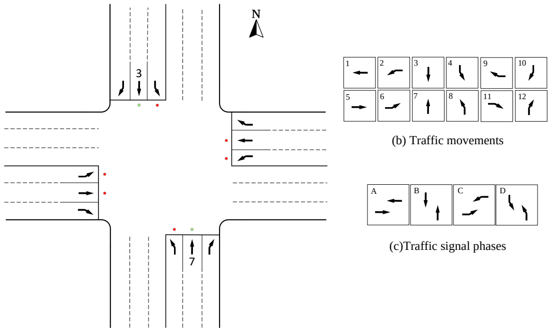
However, these sequential frameworks frequently project spatiotemporal features into unaligned high-dimensional vectors or decouple the spatial and temporal dimensions entirely. Consequently, they do not explicitly model the propagation delay of traffic waves between agents. Unlike general-purpose sequence-matching or deformable attention models (Zhu et al. 2021) that dynamically adjust receptive fields based on abstract content tokens, AlignLight proposes a physics-grounded spatiotemporal attention mechanism. By utilizing physical coordinates to modulate temporal attention distributions, the model systematically aligns its temporal focus with the physical propagation delay of incoming traffic flow prior to executing graph-level aggregation.

Preliminaries

In this section, we formally define the traffic signal control problem, abstracting the physical road network into a dynamic spatiotemporal graph and formulating the cooperative control process as a decentralized partially observable Markov decision process (Dec-POMDP).

Dynamic Spatiotemporal Graph Formulation

We model the urban road network as a directed spatiotemporal graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where nodes and edges map strictly



(a) Illustration of an intersection

(b) Traffic movements

(c)Traffic signal phases

Figure 1: Illustration of a signalized intersection. (a) Topological structure of a standard four-way intersection. (b) Twelve standard traffic movements. (c) Four typical traffic signal phases commonly used in traffic control.

to physical traffic entities.

Definition 1 (Intersections as Nodes). The set of nodes $\mathcal{V} = \{v_1, v_2, \ldots, v_N\}$ represents N signalized intersections, each controlled by an independent decentralized learning agent. Each intersection coordinates with its spatial neighbor set $\mathcal{N}_i \setminus \{i\}$, where the neighborhood size is restricted to the top-K nearest nodes ($|\mathcal{N}_i \setminus \{i\}| = K$) to adapt to irregular urban geometries.

Definition 2 (Pathways as Edges). The set of directed edges $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ denotes the physical road segments connecting adjacent intersections. We conceptualize each directed edge $e_{i,j} \in \mathcal{E}$ as a continuous geometric vector $\mathbf{g}_{i,j} = (\Delta x, \Delta y, \text{distance})$ coupled with dynamic traffic state streams, characterizing the spatial layout and directional wave propagation from intersection v_i to neighbor v_j.

Definition 3 (Signal Phases as Actions). A traffic movement $m \in \mathcal{M}$ represents vehicles traversing an intersection from an incoming approach to an outgoing approach. A local signal phase $p \in \mathcal{P}$ is defined as a permissible set of non-conflicting traffic movements executed simultaneously (as illustrated in Figure 1c), where $\mathcal{P}$ denotes the discrete phase space.

Decentralized MARL Formulation

We formulate network-level traffic signal coordination as a Dec-POMDP, defined by the multi-agent tuple $\langle \mathcal{I}, \mathcal{S}, \mathcal{A}, \mathcal{T}, \mathcal{R}, \Omega, \mathcal{O}, \gamma \rangle$. Here, $\mathcal{I} = \mathcal{V}$ represents the set of environmental agents, $\mathcal{S}$ is the global state space, and $\mathcal{A} = \prod_{i \in \mathcal{I}} \mathcal{A}_i$ is the joint action space. Multiple agents cooperate to optimize grid-wide throughput by making localized decisions based on partial historical observations $\mathbf{o}_i^t \in \Omega$.

Observation ($\mathbf{o}_i^t \in \Omega$): To capture continuous traffic flow and historical state evolution, the local observation $\mathbf{o}_i^t$ is structured as a multivariate historical sequence tensor $\mathbf{o}_i^t \in \mathbb{R}^{K \times W \times D}$, where W is the temporal sliding window length and D is the number of traffic attributes. Specifically, for each neighbor intersection $j \in \mathcal{N}_i \setminus \{i\}$ and historical step $\tau \in [t - W + 1, t]$, the localized incoming traffic feature vector is formulated as:

$$\mathbf{x}_{i,j}^{\tau} = \left[f_{i,j}^{\tau}, \mathbf{p}_{i,j}^{\tau} \right] \tag{1}$$

where $f_{i,j}^{\tau}$ denotes the macroscopic directional flow volume discharging from neighbor j toward ego-intersection i, and $\mathbf{p}_{i,j}^{\tau}$ represents the active phase encoding of the neighbor. This historical sequence provides the temporal context required to track physical traffic wave propagation.

Action ($a_i^t \in \mathcal{A}_i$): At each discrete time step t, agent i selects a local action $a_i^t \in \mathcal{A}_i \equiv \mathcal{P}$, which designates the green phase configuration to be executed for the upcoming operational interval Δt.

Reward ($r_i^t \in \mathcal{R}$): The immediate reward r_i^t evaluates localized congestion relief, defined as the negative sum of the queued vehicles across all incoming lanes $\mathcal{L}_i^{in}$ at the ego-intersection:

$$r_i^t = - \sum_{l \in \mathcal{L}_i^{in}} Q_l^t \tag{2}$$

where Q_l^t denotes the number of queued vehicles on lane l at time step t.

Objective: The primary objective for each agent i is to optimize a parameterized policy $\pi_\theta(a_i^t | \mathbf{o}_i^t)$ that maximizes the expected discounted return:

$$J(\pi_\theta) = \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t r_i^t \right] \tag{3}$$

where $\gamma \in [0, 1)$ is the discount factor balancing immediate local pressure relief with long-term network-level coordination.

Method

In this section, we detail the architecture of AlignLight. Designed for decentralized multi-agent coordination, our model improves upon standard GNN-based methods by explicitly modeling the physical propagation delays of traffic flow and utilizing continuous spatial coordinates for graph aggregation.

Temporal Sequence Encoder

To capture historical traffic trends, AlignLight maintains a sliding window of length W. As defined in Section , the observation sequence for agent i is $\mathbf{o}_i \in \mathbb{R}^{K \times W \times D}$.

To extract temporal features, we apply a 1D Average Pooling operation (kernel $= 2$, stride $= 2$) across the temporal dimension of the raw sequences and project them via a weight matrix $\mathbf{W}_{\text{feat}}$. This operation reduces high-frequency noise and aligns the sequence with the macro-level frequency of traffic signal changes. We then add a static sinusoidal positional encoding ($\mathbf{PE}_{\text{static}}$) to preserve temporal order, formulating the encoded sequence representation $\mathbf{e}_i$ as:

$$\mathbf{e}_i = \text{AvgPool1D}\left(\mathbf{o}_i\right) \mathbf{W}_{\text{feat}} + \mathbf{PE}_{\text{static}} \tag{4}$$

The resulting tensor $\mathbf{e}_i \in \mathbb{R}^{K \times T \times d_{\text{model}}}$ provides a compact representation of traffic momentum prior to graph-level routing, where $T = W/2$ denotes the compressed sequence length. Through this encoding process, the absolute environmental simulation steps $[t - W + 1, t]$ are mapped onto a relative temporal index space $\tau \in \{1, \ldots, T\}$, where $\tau = T$ represents the most recent compressed historical interval.

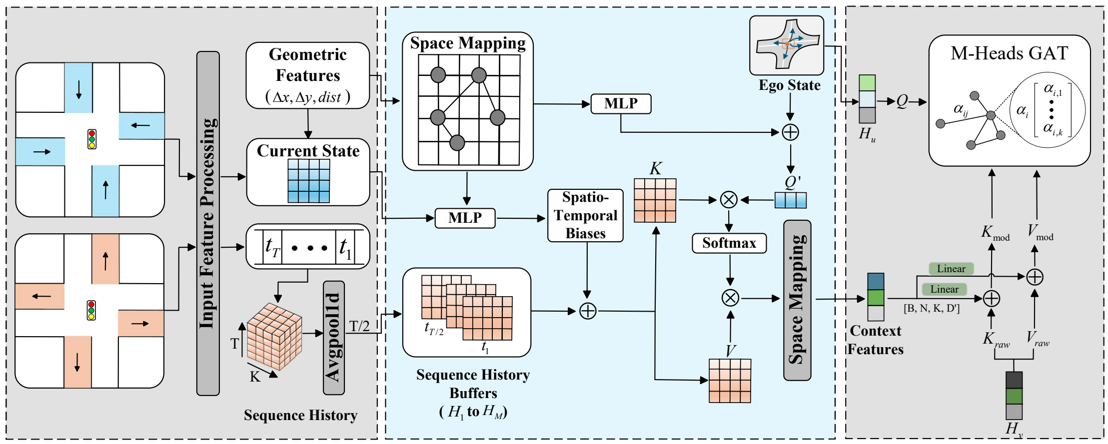


Figure 2: The overall architecture of AlignLight. The model consists of three cascaded components: (1) a temporal sequence encoder that extracts traffic flow features; (2) a phase-shifted spatiotemporal attention mechanism that uses distance-based biases to account for traffic propagation; and (3) an ego-decoupled graph aggregation module powered by continuous spatial encoding.

Spatiotemporal Attention with Physical Alignment

Traffic flow arrives at target intersections with a temporal delay governed by physical distance. To account for this, we propose a spatiotemporal attention mechanism that utilizes geometric distances to adjust temporal weights within the compressed relative index space.

First, the base query $\mathbf{q}_i$ is concatenated with physical geometric features $\mathbf{g}_{i,j} = (\Delta x, \Delta y, \text{distance})$ from neighbor j. An offset predictor infers a continuous temporal shift $\Delta_{i,j}$, representing the expected travel time from neighbor j:

$$\Delta_{i,j} = \text{Softplus}\Big(\text{MLP}\big(\mathbf{q}_i \parallel \mathbf{g}_{i,j}\big)\Big) \tag{5}$$

where $\parallel$ denotes the concatenation operation. To align the attention distribution with this physical delay, we construct a temporal bias vector $\mathbf{b}_{i,j} \in \mathbb{R}^T$, whose elements are defined across the relative sequence steps $\tau \in \{1, \ldots, T\}$:

$$\mathbf{b}_{i,j}(\tau) = -\left|(T-\tau) - \Delta_{i,j}\right| \tag{6}$$

This temporal bias is broadcasted into the attention score matrix to penalize misaligned historical states. For instance, a continuous expected delay of $\Delta_{i,j} = 3.2$ imposes a proportional L_1 penalty across the discrete relative sequence, smoothly concentrating the maximum attention density around the index $\tau = T - 3$. The aligned temporal attention output for neighbor j is then computed as:

$$\mathbf{Z}_{i,j}^{(h)} = \text{softmax}\left(\frac{\mathbf{Q}_i^{(h)}(\mathbf{K}_j^{(h)})^\top}{\sqrt{d_k}} + \mathbf{b}_{i,j}\right)\mathbf{V}_j^{(h)} \tag{7}$$

Processed via root mean square normalization (RMSNorm) (Zhang and Sennrich 2019) and residual connections, the embedding $\mathbf{h}_{i,j}$ captures the traffic state of neighbor j aligned with the physical arrival time at intersection i.

Distinction from General Relative Position Encodings. Modulating attention scores with spatial distances may structurally resemble the relative position encodings utilized in general sequence modeling (Shaw, Uszkoreit, and Vaswani 2018). However, AlignLight differs in both its generative mechanism and physical grounding. Conventional relative position encodings rely on discrete, predefined lookup tables to index static token intervals within structured grids. In contrast, AlignLight processes continuous, non-grid physical vectors $\mathbf{g}_{i,j}$ via a differentiable network. Furthermore, while general position encodings represent abstract structural distances, the generated offset $\Delta_{i,j}$ possesses explicit kinematic significance—it maps a spatial distance to an asymmetrical dynamic shift on the temporal axis, conditionally scaled by the downstream traffic pressure $\mathbf{q}_i$. This physics-grounded mapping guides the cross-attention mechanism to conform to the macroscopic laws of traffic wave propagation.

Geometry-Modulated Graph Aggregation

Standard GATs (Veličković et al. 2018) often rely on discrete adjacency matrices, which can lead to overfitting on specific topologies. AlignLight addresses this limitation by replacing discrete connections with continuous spatial positional encodings to dynamically modulate message passing.

Specifically, rather than learning abstract structural embeddings or sampling offsets, we directly utilize the absolute physical layouts of the road network. We map the deterministic geometric vector $\mathbf{g}_{i,j}$ into a high-dimensional spatial representation $\mathbf{e}_{i,j}$. To preserve stability during attention allocation, we apply RMSNorm to the base key space before edge injection, while directly augmenting the value space:

$$\begin{aligned} \mathbf{k}_{i,j}^{\text{mod}} &= \text{RMSNorm}(\mathbf{k}_j^{\text{raw}}) + \mathbf{e}_{i,j}\mathbf{W}_k^e \\ \mathbf{v}_{i,j}^{\text{mod}} &= \mathbf{v}_j^{\text{raw}} + \mathbf{e}_{i,j}\mathbf{W}_v^e \end{aligned} \tag{8}$$

To ensure that the aggregation is strictly dictated by inter-intersection physical layouts, AlignLight executes ego-decoupled aggregation by excluding the ego-state (self-loop) from this spatial message passing. By decoupling the ego-node, the model avoids topological overfitting to self-connections and forces the attention mechanism to focus entirely on incoming traffic momentum.

We define the spatial attention score $\alpha_{i,j}$ to quantify the relative importance of external physical neighbors $j \in \mathcal{N}_i \setminus \{i\}$:

$$\alpha_{i,j} = \frac{\exp\left(\mathbf{q}_i (\mathbf{k}_{i,j}^{\text{mod}})^\top / \sqrt{d_k}\right)}{\sum_{v \in \mathcal{N}_i \setminus \{i\}} \exp\left(\mathbf{q}_i (\mathbf{k}_{i,v}^{\text{mod}})^\top / \sqrt{d_k}\right)} \tag{9}$$

The finalized agent representation $\mathbf{h}'_i$ is then computed by aggregating the modulated value features using these normalized scores:

$$\mathbf{h}'_i = \sum_{j \in \mathcal{N}_i \setminus \{i\}} \alpha_{i,j} \mathbf{v}_{i,j}^{\text{mod}} \tag{10}$$

Distinction from GATv2 and Standard Graph Attentions. The ego-decoupled aggregation strategy shares conceptual commonalities with graph attention variants like GATv2 (Brody, Alon, and Yahav 2021) regarding dynamic ranking. However, our architectural motivation is driven by the causal characteristics of multi-agent traffic control rather than graph representation theory. Standard GATs and GATv2 typically enforce self-loops to stabilize node representations and mitigate over-smoothing. In localized MARL, however, including the ego-state introduces a topological confounder: the agent risks overfitting to its immediate local queues and signal phases, potentially overshadowing incoming traffic waves from adjacent neighbors. By executing strictly ego-decoupled aggregation, AlignLight establishes a physical information bottleneck. It isolates the internal state of intersection i from the incoming physical momentum of surrounding neighbors, minimizing topological confounding and enhancing zero-shot transferability across unseen networks.

Training and Optimization

AlignLight is optimized using a decentralized DQN paradigm (Mnih et al. 2015). The decoupled embedding $\mathbf{h}'_i$ is processed by a Q-network to evaluate $Q(\mathbf{h}'_i, a; \theta)$. The model is trained by minimizing the TD error $\mathcal{L}(\theta) = \mathbb{E}_{\mathcal{B}}\left[(y^t - Q(\mathbf{h}'_i, a; \theta))^2\right]$, where the target y^t is defined as:

$$y^t = r^t + \gamma \max_{a'} Q(\mathbf{h}'_i, a'; \theta^-) \tag{11}$$

Experiments

In this section, we conduct empirical evaluations on multiple real-world datasets. Our experiments are designed to assess AlignLight across three dimensions: overall performance, traffic flow efficiency, and generalization under topological shifts.

Datasets

Following established benchmarks (Zheng et al. 2019), we evaluate our framework on real-world datasets from Jinan,

Algorithm 1: Execution Procedure of AlignLight

Initialize: Replay buffer $\mathcal{D} \leftarrow \emptyset$, Q-network θ, target network $\theta^- \leftarrow \theta$.
for episode $= 1, \dots, M$ **do**
  Reset environment and observe initial state $\mathbf{s}^0$.
  for time step $t = 0, \dots, H-1$ **do**
    Encode: $\mathbf{e}_i \leftarrow$ Temporal Sequence Encoder$(\mathbf{o}_i^t)$
    Context: $\mathbf{h}_{i,j} \leftarrow$ Spatiotemporal Attention$(\mathbf{e}_i, \mathbf{g})$
    Aggregate: $\mathbf{h}'_i \leftarrow$ Ego-Decoupled Aggregation$(\{\mathbf{h}_{i,j}\}_{j \in \mathcal{N}_i \setminus \{i\}})$
    Sample local action a^t via ϵ-greedy policy derived from $Q(\mathbf{h}'_i, \cdot; \theta)$.
    Execute joint action $\mathbf{a}^t$, observe rewards $\mathbf{r}^t$ and next states $\mathbf{s}^{t+1}$.
    $\mathcal{D} \leftarrow \mathcal{D} \cup \{(\mathbf{s}^t, \mathbf{a}^t, \mathbf{r}^t, \mathbf{s}^{t+1})\}$.
    if time to update **then**
      Sample mini-batch $\mathcal{B} \sim \mathcal{D}$.
      Update θ by minimizing the TD error $\mathcal{L}(\theta)$.
    end if
  end for
  Periodically update target network $\theta^- \leftarrow \theta$.
end for

Hangzhou, and New York. As illustrated in Figure 3, these datasets represent diverse topological structures and spatial dimensions, ranging from symmetric regular intersections to large-scale asymmetric corridors. Comprehensive details regarding the physical geometric configurations (e.g., exact segment lengths, lane capacities) and dynamic traffic flow distributions are provided in Appendix . While these deterministic geometric parameters define the physical road constraints, the decentralized agents must simultaneously adapt to a vast and diverse range of unobserved peak-hour vehicle arrival patterns (Zhang et al. 2019).

Experimental Settings

We utilize the CityFlow simulator (Zhang et al. 2019) for all experiments. The decision interval is set to $\Delta t = 15$ seconds, with a 3-second yellow light and a 2-second all-red clearance time applied universally across all models to ensure physical consistency.

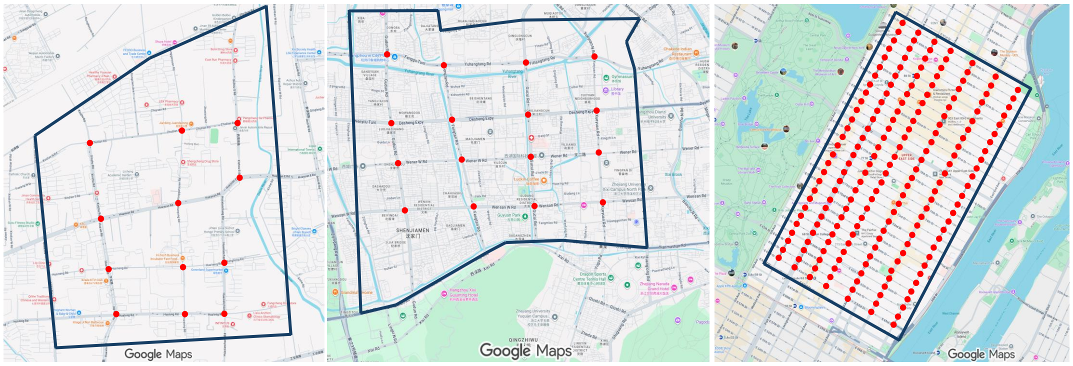
Evaluation Metrics

We adopt the average travel time (ATT) as the primary metric to measure macroscopic system efficiency. To provide a granular analysis of traffic flow, we also report network throughput (total completed vehicles) and the number of stops. All results are averaged over three independent runs with distinct random seeds.

Compared Methods

To thoroughly evaluate AlignLight, we categorize the compared baselines under three distinct operational paradigms based on their training protocols and structural definitions:

- **Online RL Baselines:** FRAP (Zheng et al. 2019) (phase competition-based); MPLight (Chen et al. 2020) (max-pressure-based RL); CoLight (Wei et al. 2019b) (graph



(a) Dongfeng Sub-district, Jinan, China

(b) Gudang Sub-district, Hangzhou, China

(c) Upper East Side, Manhattan, New York, USA

Figure 3: Topologies of the evaluated real-world road networks. (a) Jinan (3×4 grid). (b) Hangzhou (4×4 grid). (c) Manhattan, New York (28×7 grid).

attention-based communication); and PRGLight (Zhao, Hu, and Wang 2021) (max-pressure routing integrated with GNNs).

- **Offline / Sequence-Modeling Baselines:** Decision Transformer (Chen et al. 2021a) (recasting reinforcement learning as a conditional sequence generation problem) and TransformerLight (Wu et al. 2023) (utilizing gated transformer blocks to autoregressively model localized historical trajectories).
- **State-of-the-Art Topological Baselines:** Efficient-MPLight and Advanced-MPLight (Zhang et al. 2022), along with Advanced-CoLight (ADCO), which serves as the primary topology-dependent baseline for spatio-temporal MARL.

Main Results and Analysis

Table 1 presents the ATT comparison across training scenarios (Jinan, Hangzhou) and the zero-shot testing environment (New York). The evaluated models are grouped systematically by their underlying training paradigms to maintain methodological consistency.

Performance in Standard Environments. In the standard grid structures of Jinan and Hangzhou, AlignLight yields competitive optimization efficiency, matching the performance of the leading topology-dependent baseline, Advanced-CoLight. Concurrently, it systematically outperforms the sequence-modeling approaches in the majority of training scenarios. For example, in the Hangzhou 2 scenario, AlignLight reduces the ATT to 306.82s, compared to 313.91s for TransformerLight and 372.01s for Decision Transformer. This demonstrates that even in environments where spatial connectivity patterns remain fixed and can be easily memorized by structural graph models, our continuous coordinate modulation preserves high learning capability.

Zero-Shot Generalization under Topological Shifts. The primary validation of architectural robustness occurs when transitioning directly to the unseen, large-scale asymmetric corridors of New York without localized fine-tuning. As reported in Table 1, pure sequence-modeling paradigms exhibit performance degradation under topological shifts. In the New York 1 environment, Decision Transformer and TransformerLight record ATTs of 1091.11s and 1097.33s, respectively. Similarly, the standard topological baseline ADCO degrades to 970.05s due to structural overfitting to fixed discrete adjacency matrices.

In contrast, AlignLight achieves an ATT of 864.87s under identical zero-shot conditions, outperforming Advanced-CoLight by 10.84% and TransformerLight by 21.18%. This performance divergence indicates that pure sequence-matching architectures project spatiotemporal features into high-dimensional latent spaces without explicit physical alignment, losing structural fidelity under out-of-distribution transformations. By parameterizing continuous geometric coordinates to model physical propagation delays ($\Delta_{i,j}$) and executing ego-decoupled aggregation, AlignLight decouples universal kinematic regularities from specific graph layouts, establishing superior cross-city transferability.

5.5.1 Zero-Shot Generalization under Topological Shifts. The structural transferability of a multi-agent control architecture determines its applicability across heterogeneous environments. Standard node-centric graph models risk overfitting message-passing weights to fixed training topologies. To evaluate this characteristic, we deploy the model weights trained exclusively on $Hangzhou_1$ directly onto unseen environments ($Jinan$ and $NewYork$) without localized fine-tuning.

As depicted in Figure 4, ADCO exhibits notable performance degradation when confronted with the out-of-distribution configurations of $NewYork_1$ and $NewYork_2$, illustrating the limitations of memorized structural configurations under topological shifts. In contrast, AlignLight maintains stable optimization efficiency. This performance indicates that AlignLight preserves the collaborative advantages of graph-based message passing while minimizing topological overfitting through continuous coordinate modulation.

5.5.2 Training Stability and Convergence. To evaluate sample efficiency and training stability, we compare the learning curves across four traffic scenarios in Figure 5. As illus-

Table 1: Comparison of average travel time (ATT) in seconds across seven real-world traffic scenarios. Baselines are grouped systematically by their underlying operational paradigms. **Bold** indicates the best performance.

Method	Jinan			Hangzhou		New York	
	1	2	3	1	2	1	2
FRAP	299.56	268.57	269.20	308.73	355.80	1192.23	1470.51
MPLight	297.68	274.32	268.00	313.16	355.35	1321.40	1642.05
CoLight	271.17	251.22	248.87	300.07	339.76	1065.64	1367.54
Efficient-MPLight	261.81	241.35	238.80	284.49	321.08	1301.83	1485.52
Advanced-MPLight	258.29	234.78	231.76	273.26	312.68	1198.64	1439.26
AttendLight	277.53	250.29	248.82	293.89	345.72	1586.09	1683.57
PRGLight	291.27	257.52	261.74	301.06	369.98	1283.37	1472.73
Advanced-CoLight	245.73	**232.63**	**229.01**	**270.45**	310.74	970.05	1300.62
Decision Transformer	274.79	244.19	254.45	284.02	372.01	1091.11	1463.83
TransformerLight	256.16	235.78	232.39	273.18	313.91	1097.33	1413.25
AlignLight	**245.05**	233.52	229.41	271.06	**306.82**	**864.87**	**1267.92**

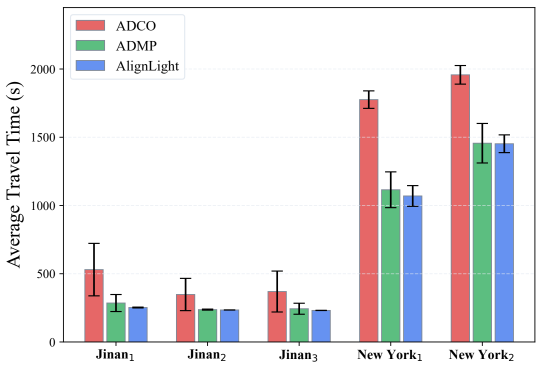


Figure 4: Zero-shot generalization performance. Models trained exclusively on $Hangzhou_1$ are evaluated on unseen networks.

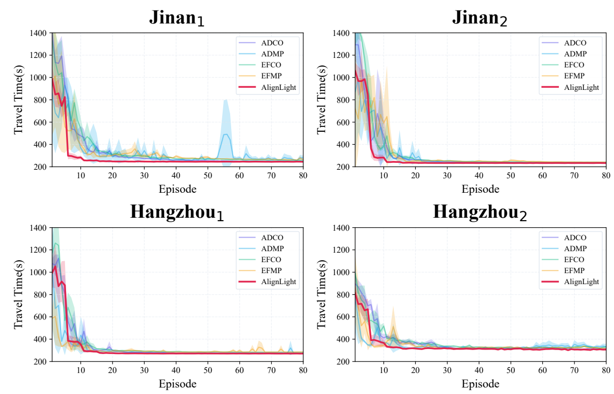


Figure 5: Training convergence curves across four datasets. AlignLight achieves a stable steady-state with reduced variance compared to the baselines.

trated, AlignLight exhibits accelerated convergence during the initial training phases relative to the baseline models. Furthermore, the learning trajectories of AlignLight reach a lower steady-state travel time with reduced variance. These profiles indicate that integrating continuous spatial coordinates into the attention mechanism stabilizes the underlying multi-agent learning dynamics. By explicitly accounting for physical propagation delays, the decentralized agents coordinate their control actions more consistently, minimizing the oscillatory behaviors frequently observed in standard MARL implementations.

Ablation Study

To isolate the architectural contributions of the proposed modules, we evaluate three variants: (1) *w/o Seq*, which restricts the historical observation window to a single snapshot ($W = 1$); (2) *w/o Joint*, which disables the joint spatiotemporal attention; and (3) *w/o Edge*, which removes the edge-injection mechanism. We evaluate these variants on $Jinan_1$, $Hangzhou_2$, and $NewYork_1$.

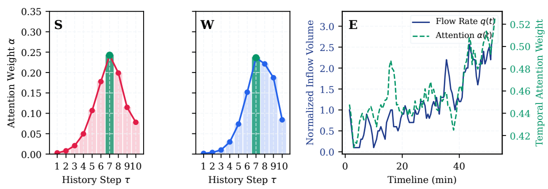
Table 2 summarizes the performance degradation observed upon removing specific components. In standard grid networks ($Jinan_1$ and $Hangzhou_2$), the *w/o Edge* variant shows the most distinct performance decline, indicating that continuous spatial coordinate injection provides a critical inductive bias for graph-level message passing under regular topologies. In contrast, for the $NewYork_1$ dataset, the *w/o Seq* variant exhibits the largest increase in ATT (from 864.87s to 969.75s). This demonstrates that in road networks characterized by longer spatial corridors, modeling temporal dynamics over a historical horizon is critical for capturing traffic propagation, as instantaneous snapshots cannot account for the delayed arrival of incoming traffic waves across extended physical distances.

Spatiotemporal Alignment Analysis

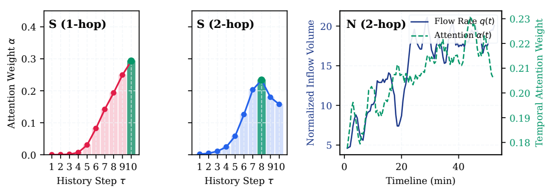
To analyze the empirical characteristics of the spatiotemporal alignment mechanism, we conduct case studies on representative intersections characterized by high traffic volumes. We visualize the learned temporal attention weights (α) distributed over the compressed historical sequence window. The compressed history length is $T = 10$, where $\tau = 10$ represents the most recent relative interval and $\tau = 1$ represents the oldest observation. Figure 6 presents these selected

Table 2: Ablation study of AlignLight components measured by ATT (s). *w/o Seq*, *w/o Joint*, and *w/o Edge* denote the removal of the sequence encoder, joint spatiotemporal attention, and edge-injection mechanism, respectively.

Method	$Jinan_1$	$Hangzhou_2$	$NewYork_1$
AlignLight	245.05	306.82	864.87
w/o Seq	245.22	309.93	969.75
w/o Joint	245.10	310.66	952.49
w/o Edge	246.21	313.78	899.65



(a) $Hangzhou_2$



(b) New $York_1$

Figure 6: Distribution of temporal attention weights over the historical sequence. (a) A standard intersection in $Hangzhou_2$, where attention peaks at $\tau = 7$. (b) A grid intersection in $NewYork_1$, showing distinct temporal attention peaks for 1-hop ($\tau = 10$) and 2-hop ($\tau = 8$) neighbors, reflecting varying propagation delays.

case studies from the $Hangzhou_2$ and $NewYork_1$ environments.

Capturing propagation delay. Figure 6a illustrates the temporal attention distribution for a standard intersection approach in Hangzhou. Rather than allocating the highest attention density to the most recent observation ($\tau = 10$), the temporal weights for the adjacent intersection systematically peak at $\tau = 7$. This distribution indicates that the model implicitly captures the physical propagation delay, identifying that traffic states from this neighbor require approximately three time steps to physically traverse the link and reach the target intersection.

Distance-aware phase shift. The correlation between physical spatial distance and temporal focus is further evidenced in the Manhattan network (Figure 6b), which is characterized by asymmetric one-way corridors and varying neighborhood lengths. For the immediate 1-hop neighbor, the attention allocation peaks at $\tau = 10$, indicating an immediate reliance on the nearest upstream traffic state. Conversely, for the more distant 2-hop neighbor, the attention peak shifts backward in the historical window to $\tau = 8$. This shift demonstrates that AlignLight dynamically modulates its temporal receptive field based on spatial distance coordinates, synchronizing its attention allocation with approaching traffic waves from distinct physical locations.

Conclusion

In this paper, we address the limitations of standard graph-based MARL architectures in traffic signal control, specifically their tendency to overfit training topologies and omit physical propagation delays. We propose AlignLight, an attention-based framework that integrates continuous geometric coordinates to model spatiotemporal physics. By modulating temporal attention via spatial distances and executing ego-decoupled graph aggregation, AlignLight synchronizes its receptive focus with actual travel times and eliminates topological confounding. Empirical evaluations confirm that AlignLight stabilizes decentralized training, achieves competitive efficiency, and delivers superior zero-shot transferability across unseen, heterogeneous networks.

In the future, we plan to extend this alignment framework to accommodate highly heterogeneous intersection layouts and systematically investigate its robustness under practical constraints, such as communication latency and sensor failures, thereby facilitating the transition toward physical urban deployments.

Implementation Details

Detailed Road Network Configurations

As discussed in the main text, to evaluate the structural robustness and zero-shot transferability of AlignLight, we utilize three real-world road networks with diverse spatial topologies. To ensure microscopic simulation fidelity within the CityFlow engine (Zhang et al. 2019), all approaches at every intersection are uniformly configured with six lanes (three incoming lanes and three outgoing lanes). The physical geometric specifications are detailed below:

- **Jinan Network (3×4 grid):** An asymmetric grid topology comprising 12 signalized intersections. The geometric distances between adjacent intersections are set to 400 meters for the East-West road segments and 800 meters for the South-North arterial segments.
- **Hangzhou Network (4×4 grid):** A standard 4×4 urban grid comprising 16 intersections. The spatial intervals are 800 meters for the East-West segments and 600 meters for the South-North segments, representing a typical symmetric urban district.
- **Manhattan, New York Network (28×7 grid):** A massive, densely packed corridor network comprising 196 intersections. Unlike the Jinan and Hangzhou networks, the New York grid utilizes a strictly symmetric and dense spatial dimension, with both East-West and South-North segments fixed at a 300-meter physical distance. This extreme-scale topology serves as our primary testbed for zero-shot generalization.

Hardware and Software Specifications

All experiments are conducted on an Ubuntu 22.04 server equipped with an AMD Ryzen 7 9700X CPU, 32 GB RAM, and an NVIDIA GeForce RTX 5070 Ti GPU (CUDA 12.8). The neural network models are implemented using Python 3.10 and PyTorch. The microscopic traffic environment is simulated via the open-source CityFlow engine (Zhang et al. 2019).

Environment and RL Hyperparameters

Table 3 summarizes the configurations for the traffic simulation environment and the reinforcement learning training process. To ensure physical realism and prevent collision during phase transitions, a strict 3-second yellow light and a 2-second all-red clearance time are universally applied across all intersections.

Table 3: Environment and Reinforcement Learning Hyperparameters.

Parameter	**Value**
Traffic Environment	
Simulation Duration	3,600 s
Decision Interval (Δt)	15 s
Yellow Light Time	3 s
All-Red Clearance Time	2 s
Observation Distance	167 m
Max Spatial Neighbors (K)	5
DQN Training	
Optimizer	AdamW
Learning Rate	0.001
Weight Decay	0.01
Discount Factor (γ)	0.8
Replay Buffer Capacity	12,000
Batch Size	20
Target Network Update Freq.	5 rounds
Exploration Strategy (ϵ)	Linear decay (0.8 $\rightarrow$ 0.2, rate=0.95)

Architecture Hyperparameters

The architectural configurations of the proposed AlignLight model are detailed in Table 4. Note that the raw observation sequences are compressed via 1D Average Pooling before being processed by the spatiotemporal attention module, ensuring macro-level frequency alignment while reducing computational overhead.

Table 4: AlignLight Architecture Hyperparameters.

Parameter	**Value**
History Sequence Window (W)	20
Hidden Dimension (d_{model})	32
Attention Heads (N_{head})	5
Transformer Encoder Layers	1
Feedforward Dimension (d_{ff})	128
Activation Function	GELU
Dropout Rate	0.1
QK Normalization	True

Additional Experimental Results

Network Throughput and Number of Stops

In the main text, we primarily report the Average Travel Time (ATT) to evaluate overall traffic efficiency. To provide a more comprehensive macroscopic analysis, we present the extended evaluations of Network Throughput (total completed vehicles) and Total Number of Stops in Table 5.

As shown in Panel A, in relatively simple scenarios (e.g., $Jinan$, $Hangzhou\_Real$), multiple baselines including AlignLight achieve identical throughput, indicating that all vehicles generated in the dataset successfully completed their trips and the network capacity is far from saturated. However, as the network scales to the extreme complexity of the $NewYork$ environments, AlignLight consistently clears more vehicles than the leading baseline (ADCO).

Crucially, as demonstrated in Panel B, AlignLight strictly dominates the Total Number of Stops metric across **all seven traffic scenarios**. Even in standard grid scenarios where overall throughput reaches the theoretical upper bound uniformly across baselines, AlignLight significantly minimizes vehicle halts. This confirms that explicitly aligning temporal attention with physical propagation delay enables the formation of smooth "green waves" along arterial roads, effectively reducing stop-and-go behaviors and mitigating potential carbon emissions.

Table 5: Extended evaluation of Network Throughput and Total Number of Stops across seven traffic scenarios. For throughput, higher is better (↑). For total stops, lower is better (↓). **Bold** indicates the best performance. Hyphens (-) denote baseline models not evaluated on the extreme-scale New York scenarios. *Abbreviations: EFCO (Efficient-CoLight), EFMP (Efficient-MPLight), ADMP (Advanced-MPLight), ADCO (Advanced-CoLight).*

Method	Jinan			Hangzhou		New York	
	Real	2000	2500	Real	5816	Double	Triple
Panel A: Network Throughput (↑)							
EFCO	**6295.00**	**4365.00**	**5494.00**	**2983.00**	5866.23	-	-
EFMP	6290.37	**4365.00**	**5494.00**	2979.30	5909.80	-	-
ADMP	6249.67	**4365.00**	**5494.00**	2974.87	5929.63	-	-
ADCO	**6295.00**	**4365.00**	**5494.00**	**2983.00**	5886.47	9209.60	11097.67
AlignLight	**6295.00**	**4365.00**	**5494.00**	**2983.00**	**5948.90**	**9290.50**	**11434.27**
Panel B: Total Number of Stops (↓)							
EFCO	16203.5	9546.1	12506.7	5871.9	17031.6	-	-
EFMP	17219.5	9596.3	12925.2	6064.1	18217.4	-	-
ADMP	13576.6	7882.2	10293.1	4115.1	15386.8	-	-
ADCO	12727.4	7843.5	10162.0	3951.6	14797.2	71801.2	73043.4
AlignLight	**12229.1**	**7565.3**	**9744.9**	**3563.9**	**12221.1**	**67643.5**	**70542.8**

Computational Cost Analysis

To evaluate the deployment feasibility of AlignLight in real-world traffic signal controllers, we analyze its computational overhead from two critical dimensions: model complexity and real-time inference latency.

Model Complexity. As shown in Table 6, although AlignLight introduces a dynamic spatiotemporal attention mechanism and continuous geometric modulation, its parameter count remains highly efficient. Under the actual microscopic simulation settings (e.g., streamlined state dimensions and 4-phase action spaces), the total parameter size of AlignLight is merely $\sim$55.8K. While this is moderately larger than the leading baseline ADCO ($\sim$19.1K) due to the incorporation of continuous spatial encoding and historical temporal sequences, the overall architecture remains extremely lightweight. By utilizing the 1D temporal pooling strategy ($W = 20 \to T = 10$) to compress feature dimensions, AlignLight achieves state-of-the-art coordination performance without incurring prohibitive memory bottlenecks.

Inference Latency. In real-world traffic signal control, the decision-making interval typically spans 10 to 15 seconds. To rigorously simulate real-time edge deployment, we benchmarked the single-step forward inference latency with a strictly sequential batch size ($B = 1$) across different network scales: standard grids ($N = 12$, Hangzhou) and massive asymmetric corridors ($N = 196$, New York). To ensure a strictly fair comparison, all models were evaluated on the same CPU environment. Despite processing dense spatiotemporal history, AlignLight achieves an average latency of 7.06 ms on the massive New York grid, which is highly competitive with simpler baselines like ADCO (5.89 ms) and significantly faster than ADMP (12.15 ms). More importantly, benefiting from modern deep learning frameworks, AlignLight can be seamlessly accelerated on edge GPUs, further reducing the latency to 2.85 ms. This millisecond-level delay is practically negligible in physical traffic systems, proving that our geometric modulation completely satisfies the stringent real-time computational constraints of edge devices.

Table 6: Comparison of computational cost. *Parameters* denotes the trainable weights. Inference latency is strictly evaluated with a batch size of 1 ($B = 1$) to simulate real-world sequential edge deployment. *Abbreviations: EFMP (Efficient-MPLight), EFCO (Efficient-CoLight), ADMP (Advanced-MPLight), ADCO (Advanced-CoLight).*

Method	**Params (K)**	**CPU Latency (ms)**		**GPU Latency (ms)**
		$\mathbf{N = 12}$	$\mathbf{N = 196}$	$\mathbf{N = 196}$
EFMP	-	1.62	1.52	-
EFCO	-	3.79	6.11	-
ADMP	-	10.80	12.15	-
ADCO	$\sim$19.1	3.51	5.89	-
AlignLight	$\sim$55.8	2.30	7.06	2.85

Limitations and Broader Impact

Limitations

While AlignLight demonstrates strong performance and generalization in complex environments, we acknowledge certain limitations that present opportunities for future research.

First, regarding traffic dynamics, the spatiotemporal attention mechanism relies on the continuous momentum of traffic flow to calculate physical propagation delays. In scenarios with extremely sparse traffic (e.g., midnight off-peak hours) or at isolated marginal intersections, the continuous traffic waves may dissipate. Under such low-pressure conditions, the model's temporal attention may degrade to a relatively uniform distribution. Although this does not negatively impact the actual travel time—as traffic pressure is inherently low—the interpretability of the distance-aware geometric modulation becomes less prominent.

Second, regarding hardware and deployment reliability, AlignLight assumes access to the macroscopic traffic states (flow and phase) from neighbors. In real-world V2X (Vehicle-to-Everything) deployments, communication latency, packet loss, or roadside sensor (e.g., radar or camera) failures could introduce missing or noisy data into the historical channel sequences. Future work could integrate attention-masking mechanisms or temporal interpolation modules to enhance the model's robustness against such hardware-induced perturbations.

Lastly, regarding topological extremes, while our geometric modulation has proven highly effective across both symmetric grids (e.g., Hangzhou) and highly asymmetric corridors (e.g., the Manhattan topology), its zero-shot transferability to extremely irregular road structures—such as complex roundabouts or five-way junctions—remains to be fully explored. In such non-orthogonal topologies, the current Cartesian relative coordinates $(\Delta x, \Delta y)$ might be less optimal than polar coordinate representations, which presents a promising direction for extending physics-informed MARL architectures.

Broader Societal Impact

Traffic congestion contributes significantly to global greenhouse gas emissions and economic inefficiency. By optimizing traffic signal control through spatiotemporal physical alignment, AlignLight substantially reduces both average travel times and vehicle stop-and-go behaviors. This smoothing of traffic flow (i.e., the formation of green waves) directly translates to lower fuel consumption and reduced carbon emissions in urban environments. Furthermore, improving intersection throughput enhances overall urban mobility and road safety.

Reproducibility Statement

To ensure the reproducibility of our research, we have detailed all environment configurations, baseline implementations, and algorithmic hyperparameters in Appendix . Furthermore, the complete, anonymized source code, including the proposed continuous spatial encoding modules, traffic simulators, and pre-trained model weights for zero-shot generalization testing, are provided in the supplementary material. Upon acceptance, the full repository will be publicly open-sourced under an MIT License to facilitate future research in geometric multi-agent reinforcement learning.

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